

0016 — ES × MenthorQ Gamma Levels: touch study, a data-corruption incident, and the CR-0DTE fade

Series: MC Setup Research Notes · **Session:** S73 night (2026-07-15) · **Instrument:** ES futures 5M RTH

CONFIDENCE: MEDIUM — one year of level history (that's all that exists), n=60 for the surviving trade rule, and bar prices repaired via a daily-offset method with ±1–2pt residual uncertainty. The negative findings (retractions) are HIGH confidence.

TL;DR: MenthorQ's daily gamma levels were tested against ES 5-minute bars over ~1 year. A mid-study audit found the continuous ES contract is **back-adjusted** (bars drift up to +465 pts from the actual prices levels are struck at), which had fabricated an apparent "negative-GEX Call-Resistance fade" edge — **retracted**, along with a 1D-Max fade. On price-repaired bars, the major levels (CR/PS) are touched too rarely or too weakly to trade, but the **0DTE levels** are dense and reactive: **fading the first approach-from-below touch of Call-Resistance-0DTE earns +\$123/trade (~60 trades/yr, +\$7.4K/yr/lot, maxDD -\$1K), positive in all 36 stop/target cells and positive out-of-sample.** Candidate — not yet validated (needs NQ replication + live paper).

1. The setup (self-contained)

- **MenthorQ levels** (options-dealer positioning analytics, computed EOD 11pm ET for futures from the option chain — so date-d levels are known before session d; causal):
 - **CR / PS** — strike with the largest net call / put gamma concentration (major wall).
 - **HVL** — gamma-regime flip line (inflection of cumulative gamma curve).
 - **CR0 / PS0 / HVL0 / GW0** — same concepts computed from 0DTE options only.
 - **1D Max / 1D Min** — options-implied expected daily move boundaries.
 - **GEX regime** — prior-EOD aggregate net gamma exposure (dealer long/short gamma), from MenthorQ gamma-insights (365-day history).
- **Bars** — ES 5M RTH continuous, **price-repaired to unadjusted front-contract values** (see §3.1) — 228–256 sessions overlapping the level history (2025-07 → 2026-07).
- **Touch semantics** (critical, user-caught): a resistance touch requires the prior 3 bars to close below the level (rally into it); support the mirror. Without this, ~44% of "touches" are approaches from the wrong side and every statistic is contaminated.

2. Question

Do price bars respect these levels in a tradeable way — raw, or conditioned on GEX regime and time of day? And do MenthorQ's own published claims hold on our data?

3. How we tested it

3.1 The data-corruption incident (read before trusting any bar-vs-level study)

data/bars/_continuous.parquet are **back-adjusted**: each quarterly roll shifts all earlier prices by the roll gap. Levels are struck in actual contract prices, so the misalignment grows with lookahead:

date	continuous-bar mid vs actual
2024-07	+462 pts
2025-07	+225–263 pts
2025-10	+166 pts
2026-01	+124 pts
2026-04	+37 pts
2026-06+	~0 (front contract)

Repair: per-session offset = continuous 15:59 close – actual front-contract close (Yahoo ES=F daily), rolling-median smoothed, subtracted from all OHLC (scripts/es_unadjust.py → _continuous_unadj.parquet, audit in data/bars/es_offsets.csv). Residual uncertainty ±1–2 pts (Yahoo close timing vs our 15:10 CT last bar).

3.2 Studies (all on repaired bars unless marked)

- **Touch/react:** touch = 5M bar range contains level ±1pt with correct approach; outcome race to ±4pt within 40 min → hold (reject) vs break.
- **Fade sim:** short (resistance) / long (support) at the touch; stop/target grid {3..10}×{6..20} pts; 1.25pt round-trip friction; ES \$50/pt; mark-to-close if neither hits.
- **Conditioning:** prior-EOD GEX sign & 1-yr percentile; hour of day; QScore; CR–HVL distance.
- **OOS:** chronological first-⅔ / last-⅓ split.

4. Results

4.1 RETRACTED (misaligned-bars artifacts — kept for the record)

Claimed edge (pre-repair)	Pre-repair result	Post-repair reality
Neg-GEX morning CR fade	n18, +\$104/tr, +\$1,875/yr, 35/36 cells +	n=1 trade/yr. Artifact.
1D-Max fade (any regime)	n102 touches, 75.5% hold, +\$131/tr	n15 touches, fade -\$79/tr
Hour-of-day hold pattern (67.9% AM / 35.3% midday)	looked strong	contaminated; not yet re-established
Gap-direction discriminator	looked predictive	dead even pre-repair on full sample

4.2 All levels — repaired bars (touch/react + fade at stop8/tgt10)

Level	Kind	Touches/yr	Hold %	negGEX hold	posGEX hold	Fade E\$/trade	Fade E\$ negGEX
CR 0DTE	res	103	81.6	84.2% (n38)	80.0% (n65)	+\$174	+\$143
PS 0DTE	sup	115	71.3	66.1% (n56)	76.3% (n59)	+\$68	+\$52
CR (major)	res	26	80.8	80.0% (n5)	81.0% (n21)	+\$255	-\$12
PS (major)	sup	22	72.7	86.7% (n15)	42.9% (n7)	+\$151	+\$198
1D Min	sup	52	75.0	73.0% (n37)	80.0% (n15)	+\$43	+\$73
1D Max	res	15	86.7	88.9% (n9)	83.3% (n6)	-\$79	-\$62

Majors confirm the S66 finding: they're rarely reached (PS 22×/yr, CR 26×/yr). The 0DTE levels hug price → 100+ touches/yr each, with high hold rates in **both GEX regimes**.

4.3 The survivor — CR-0DTE first-touch fade (short first from-below touch, any regime)

60 entries/yr, spread evenly: 2–9 per month across all 12 months (no regime clustering).

Stop/target sweep, E\$/trade (win%): **36/36 cells positive**

stop1tgt	6	8	10	12	15	20
3	+42 (57%)	+44 (47%)	+56 (42%)	+79 (40%)	+115 (38%)	+114 (33%)
4	+29 (58%)	+28 (48%)	+39 (43%)	+63 (42%)	+100 (40%)	+122 (37%)
5	+45 (65%)	+56 (57%)	+87 (53%)	+120 (52%)	+171 (50%)	+174 (43%)
6	+98 (77%)	+116 (68%)	+137 (62%)	+193 (62%)	+258 (60%)	+254 (52%)
8	+109 (82%)	+151 (77%)	+123 (63%)	+181 (63%)	+247 (62%)	+253 (55%)
10	+144 (88%)	+188 (83%)	+142 (68%)	+204 (68%)	+276 (67%)	+292 (60%)

Reference cell (stop 8 / tgt 10): **n60, win 63%, +\$123/trade, +\$7,400/yr, maxDD -\$1,000**. OOS: in-sample (first ⅔) +\$170/trade → **out-of-sample (last ⅓) +\$29/trade — degraded but positive**. Surface structure is coherent: wider stops lift win% 42→88% (rejections are wicky), consistent with dealer-hedging mechanics at 0DTE strikes (high gamma → aggressive defense on first test).

4.4 Full gauntlet on ALL levels (first-touch fades, any regime) — user-prompted extension

Every level given the identical treatment: 36-cell sweep, reference cell (stop8/tgt10), chronological ⅔/⅓ OOS split, monthly spread:

Level	Dir	n/yr	Months	Cells+	Ref E\$/tr	Ref win	Total/yr	maxDD	IS E\$	OOS E\$	Verdict
CR (major)	short	18	10/12	36/36	+\$318	83%	+\$5,725	-\$462	+\$333	+\$288	SURVIVES
CR 0DTE	short	60	11/12	36/36	+\$123	63%	+\$7,400	-\$1,000	+\$170	+\$29	SURVIVES
PS 0DTE	long	68	11/12	36/36	+\$130	66%	+\$8,862	-\$2,825	+\$158	+\$77	SURVIVES
1D Min	long	37	10/12	31/36	+\$73	59%	+\$2,688	-\$1,850	+\$150	-\$88	fragile
PS (major)	long	12	6/12	25/36	-\$12	50%	-\$150	-\$950	-\$12	-\$12	fragile
1D Max	short	11	7/12	5/36	-\$160	36%	-\$1,764	-\$2,338	-\$47	-\$462	REJECT

Three survivors, symmetric structure: fade the first test of the call walls (CR major: rare but the strongest per-trade at +\$318 with OOS +\$288 and tiny DD; CR-0DTE: the volume producer), and BUY the first test of the 0DTE put wall (PS-0DTE, the long side). Combined ≈146 trades/yr, ≈+\$19K/yr gross per lot **before overlap dedup** (CR and CR-0DTE sometimes coincide at the same strike → same-day double entry needs a rule). d1_min's in-sample +\$150 collapsing to -\$88 OOS is a textbook fragility read; 1D-Max fade confirmed dead.

4.5 Comparison sweep for the retracted CR fade (repaired) — the artifact autopsy

Same construction on the major CR with the neg-GEX filter produces 1 entry in the year (the signal effectively never fires with true prices) — the pre-repair 18 "touches" were bars shifted 100–260 pts onto an untouched level.

5. Why it works / fails

0DTE options concentrate the day's heaviest gamma near the money; dealers defending those strikes hedge aggressively on the first test, producing the reject. The effect is regime-robust (works under positive and negative aggregate GEX) because 0DTE local gamma dwarfs the aggregate book at that strike, intraday. Majors fail as trades not because they don't hold (they

do, ~80%) but because they're **rarely reached** — a level that triggers 26 times a year with wide dispersion can't carry a strategy alone. 1D-Max fades fail because reaching the implied-move ceiling requires a strong trend day — exactly when resistance breaks.

6. Recommendation

Advance the THREE surviving first-touch fades to live paper trading as one suite (1 MES/ES lot each, stop 8 / target 10, first correct-side touch per level per day, journal every trade): **short CR-major, short CR-0DTE, long PS-0DTE**. Dedup rule needed before go-live: if CR and CR-0DTE coincide within ~5 pts, take ONE entry (treat as CR-major, the stronger edge). **Do not size up until:** (a) NQ replication passes, (b) ≥ 20 live paper samples per rule roughly match backtest, (c) MentorQ-definition cross-check done. 1D-Min stays on the watchlist (fragile, OOS-negative); 1D-Max fade and PS-major are rejected as standalone trades.

7. Caveats

- **One year of level history** — that is all that exists anywhere; no way to extend backward.
- Bar repair carries $\pm 1\text{--}2\text{pt}$ residual error $\rightarrow$ touch detection at $\pm 1\text{pt}$ tolerance can misclassify borderline touches; direction of bias unknown.
- Level-date causality verified for the majors (dashboard morning levels = QUIN same-date row); assumed identical for 0DTE levels (same 11pm ET compute) — verify live.
- Hold-rate metric is definition-sensitive (base rate swings 52–68% across MOVE/LOOK); the P&L sim is the number that matters.
- MentorQ subscribers watch these same levels — crowding effects unknowable from this data.
- $n=60$; a 1-year sample contains one regime cycle at most.

8. Reproduce

```
.venv/Scripts/python.exe scripts/es_unadjust.py      # bar repair (run first)
.venv/Scripts/python.exe scripts/mr_es_all_levels.py # 4.2
.venv/Scripts/python.exe scripts/es_cr0_sweep.py    # 4.3
.venv/Scripts/python.exe scripts/es_cr_fade_sweep.py # 4.4 autopsy
```

Data: data/mentorq/levels_history.csv, levels0_history.csv, gex_insights_ES1.csv, data/bars/_continuous_unadj.parquet (+ es_offsets.csv audit).